

TABLE II
Soldering Stations from Different Vendors

Features and specifications	Hakko FX-888D	Okinternational MFR-1120	Weller WX 1	Goot Japan RX-802AS
Analogue/digital	Digital	NA	Digital	Digital
Wattage				
Overall	70W	70W	200W	80W
Soldering pencil	65W (26V)	Output 60W Max at 22°C	40W	70W
Temperature range	200 - 480°C	Fixed temperature system that varies power to match the thermal demand of the device being soldered	50°C to 550°C	50°C to 450°C
Temperature stability	±1°C at idle temperature	±1.1°C in still air	±2°C	3°C
Tip to ground potential	< 2 mV	< 2 mV true RMS 50 to 500 Hz		
Tip to ground resistance	< 2 ohms	< 2 Ohms		
Thermal recovery	20 sec to reach 350°C			Six seconds to reach set temperature
Input/output voltage	220/26V AC	102 to 240V Input	230V, 50 Hz Input	220-240V input
Heating element	Ceramic heater	Proprietary skin effect heating/Curie point control		Nichrome high grade
Iron stand	Yes	Yes (WS2)	Yes	Yes
Dimensions	100×120×120mm	91×142×197mm	170×151×130mm	146×115×98mm
Cord length	1.2m	1.2m burnproof ESD safe		1.2m
Standard tip	Shape-B (No.T18-B)	SXP series cartridge tips	RT3 (0054460399)	Pencil type (user can choose)
Cleaning wire	Yes	Yes (bronze sponge)	Yes	
Cleaning sponge	Yes	Yellow wet sponge provided with the stand		Yes

3. ESD-safe equipment like soldering iron

4. ESD-safe brush

5. ESD-safe trays, bins and cabinets
A bench-top ioniser can be used to neutralise ESDs at the workstation. This is the only ESD-control method available to neutralise ESDs on essential insulators

or isolated conductors that may be at the workstation. The required limit according to ANSI/ESD S20.20 is less than ±50V offset voltage (balance). In addition to that, the discharge time to reduce +1000V to +100V and to reduce -1000V to -100V should also be measured. Faster the static elimination time, the better it is. Do look for one that strictly complies with the ANSI/ESD S20.20 standard. Refer Table I for anti-static components available from different vendors.

Correct and good-quality tools

It is easy to get away with cheap tools available in the market, but they may reduce the efficiency and quality of the

job. Select each tool very carefully, considering your requirement in detail, to effectively set up the rework station. It is best if you can use all ESD-safe equipment and tools.

The first tool that you need on your rework station is the soldering iron. Do not go for cheap plug-in type unregulated soldering irons, as these irons do not have any kind of thermal regulation. They are directly plugged into the wall socket. Such soldering irons heat up to high temperatures, so the initial soldering occurs at way too high temperatures, which leads to bad-quality soldering and possible damage to the components. With a temperature-controlled iron, you can set the temperature to be beyond the melting point of your specific solder, but still the temperature will remain within the safe limits for the component. Especially with SMD packages, using a too high temperature or holding the iron too long in a static position (because of inefficient heat transfer) may damage it.

So you should never buy cheap unregulated soldering irons, as they prove to be more expensive than the temperature-controlled irons because of the damage that they may cause.

Now, with this point being concluded, see Table II to check out some soldering stations from different manufacturers.

After a soldering station is decided, you need de-soldering equipment. De-soldering equipment can be divided into different types. The first type is for desoldering surface mount device (SMD) components, for which you need a de-soldering tool called 'hot air rework system'. As the name suggests, this tool blows hot air on the component, melting the solder on all the sides together. Once the solder has melted, you can remove the component using a pair of tweezers. Another tool called 'heated tweezers' is exactly what the



Fig. 2: Soldering station